

JANICE BENITA F

AI/ML Engineer | Explainable AI | Computer Vision | AI Agents

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PROFESSIONAL SUMMARY

Final-year Information Technology student with hands-on experience in machine learning, computer vision, explainable AI, data analytics, and full-stack AI product development. Experienced in building deployable AI systems, engineering decision-support platforms, research prototypes, and autonomous agent workflows.

EXPERIENCE

Machine Learning and Data Analytics Intern - YuvaIntern

Mar 2026 - May 2026

- Performed exploratory data analysis, feature engineering, predictive modelling, and visualisation on agricultural datasets using Python and Pandas.
- Built reusable machine-learning workflows and communicated findings through visual analysis.

Data Analytics and Machine Learning Intern - Larsen & Toubro, Chennai

May 2025 - Jul 2025

- Developed a ResNet-18 and Grad-CAM computer-vision pipeline for automated concrete-crack classification.
- Achieved 95.7% classification accuracy and analysed more than 10,000 inspection records using Python and SQL.
- Reduced inspection turnaround time from 25-30 days to 3-5 days.
- Developed dashboards and analytical workflows for infrastructure quality assessment.

Web Development Intern - CodeBind Technologies

Jul 2024 - Aug 2024

- Developed responsive web applications using HTML, CSS, and JavaScript.
- Integrated REST APIs and improved frontend usability.

SELECTED PROJECTS

SentinelOps - Autonomous AI Reliability Engineer

Evidence-first AI agent for incident investigation and bounded repair with a 20-state persisted workflow, deterministic verification, Local and Docker sandbox modes, and a human approval gate. Python, FastAPI, React, TypeScript, SQLite, Pytest, Docker.

EndoXAI - Explainable AI for Root Canal Treatment Prediction

Clinical decision-support prototype using deep-learning inference and Grad-CAM visualisation; deployed on Hugging Face Spaces. AI/ML Software Developer and Co-Author; Second Prize at IDDC 2026.

Digital Quality Intelligence Platform

Modular engineering quality decision-support platform covering Six Sigma analytics, Cp/Cpk evaluation, anomaly detection, supplier benchmarking, ACI compliance, and executive dashboards.

Explainable Concrete Crack Detection

ResNet-18 and Grad-CAM infrastructure inspection system with 95.7% classification accuracy; associated research published in IJERT.

RESEARCH

Design and Implementation of the Digital Quality Intelligence Platform: A Modular Decision-Support Software Framework for Engineering Quality Intelligence

Accepted - International Journal of Advanced Trends in Computer Science and Engineering, Vol. 15, No. 4, 2026

Explainable Deep Learning Framework for Automated Concrete Crack Detection Using ResNet-18

Published - International Journal of Engineering Research and Technology, 2026

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, SQL, HTML, CSS
AI and Machine Learning	Machine Learning, Deep Learning, Computer Vision, Explainable AI, Predictive Analytics, Feature Engineering, Model Evaluation, AI Agents
Frameworks and Libraries	PyTorch, scikit-learn, OpenCV, FastAPI, Flask, Streamlit, Pandas, NumPy, Plotly
Data and Tools	MySQL, SQLite, Power BI, Tableau, Git, GitHub, GitHub Actions, Docker, REST APIs, Hugging Face Spaces, Render, GitHub Codespaces

EDUCATION

Bachelor of Technology in Information Technology

St. Joseph's College of Engineering | 2023 - 2027 | CGPA: 8.30 / 10.00

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning

AWARDS

- Second Prize - International Digital Dentistry Congress 2026, Poster Presentation, Postgraduate Category; AI/ML Software Developer and Co-Author.
- Runner-up - CodeVerse Inter-Collegiate Technical Symposium.

CERTIFICATIONS

- Python for Computer Vision with OpenCV and Deep Learning - Udemy
- Data Science and Machine Learning Bootcamp - Udemy
- Tata GenAI Powered Data Analytics Job Simulation - Forage
- Deloitte Data Analytics Job Simulation - Forage
- Software Testing - NPTEL